Oracle AI Database 26ai

True Cache — DBA Setup Guide

Step-by-step configuration for on-premises Linux x86-64 (Release 23.26.1+)

Overview

True Cache is a diskless, read-only Oracle instance that maintains a transactionally consistent replica of the primary database in memory using Redo Apply. Unlike Redis or Memcached, True Cache is fully ACID-compliant — applications see a consistent snapshot without any manual cache invalidation.

> *Key benefit: The JDBC driver automatically routes read-only transactions to True Cache with zero application code changes. True Cache serves queries from its in-memory buffer cache; blocks not yet cached are fetched on demand from the primary.*

How True Cache differs from Active Data Guard

Feature	True Cache	Active Data Guard
Primary purpose	Read offload / low latency	HA failover + read offload
Storage on standby	None (diskless)	Full ASM disk copy
Failover capable	No	Yes
ACID consistent	Yes	Yes
License	Active Data Guard option	Active Data Guard option
Auto JDBC routing	Yes	Yes (read-only service)

Prerequisites

Software requirements

Requirement	Detail	Status
Oracle DB version	Oracle AI Database 26ai (23.26.1+) on primary	Required
True Cache binary	Same Oracle version as primary	Required
License	Active Data Guard option (included in EE)	Required
Grid Infrastructure	Oracle GI 26ai on True Cache host	Recommended
JDBC driver	Oracle JDBC 26ai thin driver	Required for auto-routing

Primary database requirements

Requirement	Detail	Status
ARCHIVELOG mode	Must be enabled — check with SELECT LOG_MODE FROM V$DATABASE	Required
FORCE LOGGING	Ensures all changes captured in redo, no unlogged direct writes	Required
DB_UNIQUE_NAME	Primary must have a unique name set in SPFILE	Required
Redo transport	ASYNC or SYNC — True Cache uses standard redo shipping	Required
Password file	orapw file required for redo transport authentication	Required

True Cache host requirements

Requirement	Detail	Status
Disk storage	NO datafiles needed — True Cache is entirely diskless	Key feature
Memory (SGA)	Size SGA to hold the read working set; larger = higher hit rate	Size appropriately
Network	Low-latency link to primary (10 GbE minimum recommended)	Required
OS / kernel	Same OS version as primary (Linux x86-64 for on-prem GA)	Required
Oracle Home	Full Oracle DB software install (same version as primary)	Required

Phase 1 — Configure the Primary Database

> *All commands in this phase run on the PRIMARY database host as SYSDBA unless noted otherwise.*

Step 1 Verify and enable ARCHIVELOG mode

Check current mode. If NOARCHIVELOG, a database restart is required.

```
-- Check current archive log mode
SELECT LOG_MODE FROM V$DATABASE;

-- If NOARCHIVELOG, enable it (requires restart):
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
ALTER DATABASE ARCHIVELOG;
ALTER DATABASE OPEN;

-- Verify
SELECT LOG_MODE FROM V$DATABASE;
-- Expected: ARCHIVELOG
```

Step 2 Enable FORCE LOGGING

Ensures no direct-path writes bypass redo generation. Critical for consistent replication.

```
ALTER DATABASE FORCE LOGGING;

-- Verify
SELECT FORCE_LOGGING FROM V$DATABASE;
-- Expected: YES
```

Step 3 Set DB_UNIQUE_NAME (if not already set)

```
ALTER SYSTEM SET DB_UNIQUE_NAME='PRODDB' SCOPE=SPFILE;
-- Requires DB restart to take effect if changed
```

Step 4 Configure redo transport to True Cache

Replace TRUECACHE1 with your chosen DB_UNIQUE_NAME for the True Cache instance. Replace truecache-host with the actual hostname or IP.

```
-- Add True Cache as redo transport destination
ALTER SYSTEM SET LOG_ARCHIVE_DEST_2=
```

```
  'SERVICE=TRUECACHE1
   ASYNC
   VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
   DB_UNIQUE_NAME=TRUECACHE1'
  SCOPE=BOTH;

ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=ENABLE SCOPE=BOTH;

-- Register both DBs in DG_CONFIG
ALTER SYSTEM SET LOG_ARCHIVE_CONFIG=
  'DG_CONFIG=(PRODDB,TRUECACHE1)' SCOPE=BOTH;
```

Step 5 Create and distribute the password file

Run on the PRIMARY OS as oracle user. Then copy to the True Cache host.

```
# On primary OS (oracle user)
orapwd file=$ORACLE_HOME/dbs/orapwPRODDB \
        password=SysPassword123 \
        entries=10 \
        format=12.2

# Copy password file to True Cache host
scp $ORACLE_HOME/dbs/orapwPRODDB \
    oracle@truecache-host:$ORACLE_HOME/dbs/orapwTRUECACHE1
```

Step 6 Add True Cache TNS entry on primary

Edit $ORACLE_HOME/network/admin/tnsnames.ora on the PRIMARY host.

```
TRUECACHE1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL=TCP)
               (HOST=truecache-host.example.com)
               (PORT=1521))
    (CONNECT_DATA =
      (SERVICE_NAME=TRUECACHE1)))
```

Phase 2 — Create the True Cache Instance

> *All commands in this phase run on the TRUE CACHE HOST unless noted otherwise.*

Step 7 Install Oracle software on True Cache host

Install the same Oracle DB 26ai binaries as the primary. Do NOT run dbca to create a normal database — you will create a True Cache instance in Step 9.

- Run the Oracle Universal Installer (OUI) in silent or GUI mode
- Install to the same ORACLE_HOME path as primary (recommended for consistency)
- Apply the same Oracle patches as the primary

Step 8 Create the seed init.ora on True Cache host

Create $ORACLE_HOME/dbs/initTRUECACHE1.ora with the following content. DBCA will expand it — this is a minimal seed only.

```
# $ORACLE_HOME/dbs/initTRUECACHE1.ora
DB_NAME=PRODDB
DB_UNIQUE_NAME=TRUECACHE1
TRUE_CACHE=TRUE

# Size SGA to hold your read working set
SGA_TARGET=8G
SGA_MAX_SIZE=8G

# Redo source
FAL_SERVER=PRODDB
FAL_CLIENT=TRUECACHE1

LOG_ARCHIVE_CONFIG='DG_CONFIG=(PRODDB,TRUECACHE1)'
LOG_ARCHIVE_DEST_1=
  'SERVICE=PRODDB
   VALID_FOR=(STANDBY_LOGFILES,STANDBY_ROLE)
   DB_UNIQUE_NAME=PRODDB'
```

Step 9 Add primary TNS entry on True Cache host

Edit $ORACLE_HOME/network/admin/tnsnames.ora on the TRUE CACHE HOST.

```
PRODDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL=TCP)
               (HOST=primary-host.example.com)
               (PORT=1521))
    (CONNECT_DATA =
      (SERVICE_NAME=PRODDB)))
```

Step 10 Run DBCA to create the True Cache instance

> *This is the key Oracle 26ai command. The -createTrueCache flag automates all wiring — no manual standby redo log creation, no RMAN restore needed.*

```
dbca -silent \
  -createTrueCache \
  -gdbName PRODDB \
  -sid TRUECACHE1 \
  -sourceDBConnectionString primary-host:1521/PRODDB \
  -sysDBAUserName sys \
  -sysDBAPassword SysPassword123 \
  -trueCacheSid TRUECACHE1 \
  -trueCacheServiceName tc_service \
  -initParams "sga_target=8g,sga_max_size=8g"
```

DBCA will perform the following automatically:

- Connect to the primary and retrieve the database metadata
- Create the True Cache Oracle instance (no datafiles created)
- Configure redo apply to stream from the primary
- Register the tc_service on the primary and True Cache
- Start Redo Apply — True Cache begins populating its buffer cache

Step 11 Verify True Cache is running and in sync

```
-- On TRUE CACHE instance
SELECT STATUS, DATABASE_ROLE, OPEN_MODE FROM V$DATABASE;
-- Expected: OPEN / PHYSICAL STANDBY / READ ONLY WITH APPLY

-- Check apply lag (should be milliseconds)
SELECT NAME, VALUE, UNIT
FROM   V$DATAGUARD_STATS
WHERE  NAME IN ('apply lag','transport lag');

-- On PRIMARY — confirm True Cache is connected
SELECT DB_UNIQUE_NAME, STATUS, GAP_STATUS
FROM   V$DATAGUARD_STATS
WHERE  DB_UNIQUE_NAME = 'TRUECACHE1';
```

Phase 3 — Create the True Cache Service

> *Run these commands on the PRIMARY database. This service is what applications connect to for automatic routing.*

Step 12 Create and start the True Cache service

```
-- On PRIMARY (as SYSDBA)
EXEC DBMS_SERVICE.CREATE_SERVICE(
  service_name       => 'tc_service',
  network_name       => 'tc_service',
  true_cache_service => 'tc_service'
);

EXEC DBMS_SERVICE.START_SERVICE('tc_service');

-- Verify
SELECT NAME, TRUE_CACHE_SERVICE, NETWORK_NAME
FROM   DBA_SERVICES
WHERE  TRUE_CACHE_SERVICE IS NOT NULL;
```

Step 13 Make the service start automatically after restarts

Register the service as a trigger so it starts on DB open.

```
-- Create a trigger to auto-start the service
CREATE OR REPLACE TRIGGER start_tc_service
  AFTER STARTUP ON DATABASE
BEGIN
  DBMS_SERVICE.START_SERVICE('tc_service');
END;
/
```

Phase 4 — Application Connection

> *Zero application code changes are required. The Oracle JDBC thin driver (26ai) automatically detects read-only transactions and routes them to True Cache via the tc_service.*

JDBC connection string (Java)

```
// Connect to True Cache service — JDBC auto-routes read vs write
String url =
  "jdbc:oracle:thin:@truecache-host:1521/tc_service";

Connection conn = DriverManager.getConnection(
```

```
    url, "appuser", "apppassword");

// Read-only transactions automatically served from True Cache
// Write transactions transparently routed to primary
```

Universal Connection Pool (UCP)

```
PoolDataSource pds = PoolDataSourceFactory.getPoolDataSource();
pds.setConnectionFactoryClassName(
    "oracle.jdbc.pool.OracleDataSource");
pds.setURL("jdbc:oracle:thin:@truecache-host:1521/tc_service");
pds.setUser("appuser");
pds.setPassword("apppassword");
pds.setInitialPoolSize(5);
pds.setMaxPoolSize(20);

// UCP automatically routes:
//   Read-only transactions  -> True Cache
//   Write transactions      -> Primary
```

Python (python-oracledb)

```
import oracledb

conn = oracledb.connect(
    user='appuser',
    password='apppassword',
    dsn='truecache-host:1521/tc_service'
)

cursor = conn.cursor()
cursor.execute(
    'SELECT * FROM customers WHERE region = :r',
    r='APAC'
)  # Served from True Cache buffer cache
```

Force read-only routing explicitly (SQL)

```
-- Pin a session to True Cache (useful for batch read jobs)
ALTER SESSION SET TRUE_CACHE_SERVICE = 'tc_service';

-- Statement-level hint
SELECT /*+ TRUE_CACHE */ *
FROM   orders
WHERE  status = 'SHIPPED';
```

Phase 5 — Monitoring & Ongoing Management

Key monitoring queries

True Cache hit ratio

Target: > 80%. Low hit ratio means SGA is too small — increase SGA_TARGET.

```
-- Run on TRUE CACHE instance
SELECT METRIC_NAME, VALUE
FROM   V$TRUE_CACHE_STATS
ORDER  BY METRIC_NAME;

-- Key metrics:
--   TRUE_CACHE_HIT_RATIO       (target > 80%)
--   BLOCKS_SERVED_FROM_CACHE
--   BLOCKS_FETCHED_FROM_PRIMARY
--   TOTAL_QUERIES_SERVED
```

Apply lag

True Cache should lag primary by milliseconds only. Alert if lag exceeds your application SLA.

```
-- Run on PRIMARY
SELECT NAME, VALUE, UNIT, TIME_COMPUTED
FROM   V$DATAGUARD_STATS
WHERE  NAME IN ('apply lag','transport lag')
AND    DB_UNIQUE_NAME = 'TRUECACHE1';

-- Run on TRUE CACHE
SELECT CURRENT_SCN, APPLIED_SCN,
       (CURRENT_SCN - APPLIED_SCN) AS SCN_LAG
FROM   V$DATABASE;
```

Buffer cache pressure

```
-- Buffer cache hit rate on True Cache
SELECT 1 - (SUM(physical_reads) /
           (SUM(db_block_gets) + SUM(consistent_gets)))
         AS buffer_hit_ratio
FROM   V$BUFFER_POOL_STATISTICS;

-- Resize SGA on the fly if needed
ALTER SYSTEM SET SGA_TARGET=16G SCOPE=BOTH;
```

Lifecycle commands

Action	Command
Stop True Cache	`SHUTDOWN IMMEDIATE; (on True Cache host)`

Action	Command
Start True Cache	`STARTUP; (on True Cache host)`
Check status	`SELECT STATUS, DATABASE_ROLE FROM V$DATABASE;`
Suspend redo apply	`ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;`
Resume redo apply	`ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;`
Remove True Cache	`dbca -silent -deleteTrueCache -sid TRUECACHE1 ...`
Disable dest on primary	`ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=DEFER SCOPE=BOTH;`

Quick Reference — Setup Checklist

Use this checklist to track progress during setup. All 13 steps must be completed in order.

	Phase	Step
[ ]	Phase 1 — Primary	Step 1: Verify / enable ARCHIVELOG mode
[ ]	Phase 1 — Primary	Step 2: Enable FORCE LOGGING
[ ]	Phase 1 — Primary	Step 3: Set DB_UNIQUE_NAME
[ ]	Phase 1 — Primary	Step 4: Configure LOG_ARCHIVE_DEST_2 for True Cache
[ ]	Phase 1 — Primary	Step 5: Create and distribute password file
[ ]	Phase 1 — Primary	Step 6: Add True Cache TNS entry on primary
[ ]	Phase 2 — True Cache host	Step 7: Install Oracle 26ai software
[ ]	Phase 2 — True Cache host	Step 8: Create seed init.ora with TRUE_CACHE=TRUE
[ ]	Phase 2 — True Cache host	Step 9: Add primary TNS entry on True Cache host
[ ]	Phase 2 — True Cache host	Step 10: Run dbca -createTrueCache
[ ]	Phase 2 — True Cache host	Step 11: Verify STATUS and apply lag
[ ]	Phase 3 — Primary	Step 12: Create and start tc_service via DBMS_SERVICE
[ ]	Phase 3 — Primary	Step 13: Create auto-start trigger

After Step 13: Connect your application to truecache-host:1521/tc_service. The JDBC/UCP driver handles automatic routing — no further changes needed.

True Cache vs Active Data Guard Standby — Step Comparison

True Cache setup is deliberately modelled on Physical Standby setup — the prerequisites and redo transport wiring are almost identical. This section maps every standby step to its True Cache equivalent so experienced DBA standby builders can ramp up instantly.

> *Green = identical step | Amber = similar but different | Red = not needed in True Cache | Blue = True Cache only*

#	Step	Match	Active Data Guard Standby	True Cache
On PRIMARY — prepare the source database				
1	**ARCHIVELOG mode**	Identical	Required. ALTER DATABASE ARCHIVELOG;	Required. Same command, no difference.
2	**FORCE LOGGING**	Identical	Required. ALTER DATABASE FORCE LOGGING;	Required. Same command.
3	**DB_UNIQUE_NAME**	Identical	Set on primary SPFILE. e.g. PRODDB	Same requirement, same parameter.
4	**Standby redo logs on primary**	Different	Required for fast failover. ALTER DATABASE ADD STANDBY LOGFILE …	Not required. True Cache does not use standby redo logs.
5	**LOG_ARCHIVE_DEST_n**	Identical	SERVICE= pointing to standby, ASYNC or SYNC.	Same syntax — SERVICE= pointing to True Cache.
6	**LOG_ARCHIVE_CONFIG DG_CONFIG**	Identical	DG_CONFIG=(PRODDB,STANDBY1)	DG_CONFIG=(PRODDB,TRUECACHE1). Same syntax.
7	**Password file**	Identical	orapwd on primary, scp to standby host.	Same — scp orapw file to True Cache host.
8	**TNS entries**	Identical	Add standby TNS alias in primary tnsnames.ora.	Add True Cache TNS alias. Same format.
On STANDBY / TRUE CACHE HOST — create the instance				
9	**Oracle software install**	Identical	Same Oracle version + patches as primary.	Same — same version, same patches.
10	**RMAN backup / restore**	Not needed	Required. RMAN DUPLICATE FROM ACTIVE DATABASE to seed standby datafiles. Time-consuming.	NOT needed. No datafiles. Buffer cache built lazily on first query from primary.
11	**Standby redo logs on standby**	Not needed	Required. Must match primary log size, count +1 group per thread.	NOT needed. Skipped entirely.
12	**init.ora / SPFILE**	Different	Standard params: DB_UNIQUE_NAME, FAL_SERVER, LOG_ARCHIVE_DEST, DB_FILE_NAME_CONVERT.	Must include TRUE_CACHE=TRUE. No DB_FILE_NAME_CONVERT. Focus on SGA_TARGET.
13	**Instance creation method**	Different	STARTUP NOMOUNT + RMAN DUPLICATE, or DBCA duplicate wizard.	dbca -createTrueCache — one command, no RMAN step.
14	**Redo Apply startup**	Different	ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;	Started automatically by DBCA. No manual command needed.

#	Step	Match	Active Data Guard Standby	True Cache
15	**Open database**	Different	ALTER DATABASE OPEN READ ONLY; then start MRP process.	DBCA handles this. Opens automatically in READ ONLY WITH APPLY mode.
Service & application configuration				
16	**Read service creation**	Different	DBMS_SERVICE.CREATE_SERVICE with standard params. App must explicitly use standby URL.	DBMS_SERVICE.CREATE_SERVICE with TRUE_CACHE_SERVICE param. JDBC auto-routes.
17	**Application changes required**	Different	App needs separate connection pool or URL for standby reads. Code change often needed.	Zero application changes. JDBC thin driver auto-routes read-only transactions.
18	**True Cache only — hit ratio monitoring**	TC only	Not applicable.	Monitor V$TRUE_CACHE_STATS — TRUE_CACHE_HIT_RATIO target > 80%.
Ongoing management differences				
19	**Failover / switchover**	Different	Full Data Guard failover and switchover. Can become primary.	No failover. Read-offload only. Cannot become primary.
20	**Disk storage to manage**	Different	Full ASM disk group — same size as primary. Monitor space, RMAN backups.	No disk. Monitor SGA/buffer cache hit ratio and memory pressure only.
21	**RMAN backups on standby**	Different	Common pattern — offload RMAN backups to standby to reduce primary I/O.	No backups. True Cache holds no persistent data.
22	**Apply lag monitoring**	Identical	V$DATAGUARD_STATS — apply lag, transport lag.	Same views. Same alert thresholds.

At a glance — what True Cache removes from a standby setup

Physical Standby requires (extra steps)	True Cache — these steps are gone
• RMAN DUPLICATE to seed all datafiles • Standby redo log creation on both hosts • ASM disk group sizing and provisioning • STARTUP NOMOUNT / MOUNT sequence • MRP process management • Separate app connection URL / pool for reads • RMAN backup jobs on standby	• **No RMAN step — buffer cache built lazily** • **No standby redo logs on either host** • **No disk storage to provision or manage** • **DBCA -createTrueCache does everything** • **No MRP — apply starts automatically** • **JDBC auto-routes — zero app changes** • **No backups — True Cache is stateless**

> *Key takeaway: If you have built a Physical Standby before, you already know 60% of True Cache setup. The primary-side configuration is identical. True Cache removes*

```
the three most time-consuming standby tasks: RMAN restore, standby redo log
creation, and manual MRP startup.
```